

LAMP-Merge: Long-tail-Aware Medical Prototype Merging for One-shot Asynchronous Clinical Models

Anonymous submission

Abstract

Multicenter medical models are commonly trained under asynchronous hospital workflows, heterogeneous acquisition devices, and incomplete diagnostic-label coverage. We study single-shot asynchronous medical model merging, where each client uploads only once after local training and the server constructs a global model without accessing raw medical images or performing federated optimization rounds. We argue that directly transferring general-purpose model merging to this setting breaks a key assumption: each local checkpoint is not a complete view of the global diagnostic space, but a partial expert supported by locally observed diseases. This mismatch leads to aggregation-induced prediction collapse, where the merged model concentrates predictions on a few dominant classes. More subtly, long-tailed medical prevalence can make such collapse deceptively competitive under raw accuracy. To address these issues, we propose LAMP-Merge, a long-tail-aware medical prototype-merging framework that shifts the merge target from global parameter interpolation to class-level diagnostic-evidence reconstruction. LAMP-Merge builds a diagnostic prototype head from client-side class statistics and calibrates disease prevalence with a bounded prior bias. Across five medical datasets, four vision-model families, three client-population sizes, and three non-IID skew levels, LAMP-Merge matches or surpasses the strongest non-LAMP baseline in 57 of 60 client-averaged settings, while improving balanced accuracy, macro-F1, and prediction-collapse diagnostics.

Index Terms: medical model merging, asynchronous fusion, predictive collapse, long-tail calibration, privacy-preserving learning

Introduction

Federated learning (McMahan et al. 2017; Li et al. 2020; Karimireddy et al. 2020; Kairouz et al. 2021) and model merging (Izmailov et al. 2018; Wortsman et al. 2022; Ilharco et al. 2023; Yadav et al. 2023) provide two important collaboration routes for privacy-sensitive multicenter medical intelligence. Federated learning enables hospitals to train collaboratively without sharing raw data (Sheller et al. 2020; Rieke et al. 2020), while model merging further reduces communication and optimization costs by integrating independently trained local models after training (Ainsworth, Hayase, and Srinivasa 2023; Stoica et al. 2024; Matena and Raffel 2022). These paradigms are particularly attractive for

medical imaging: a single hospital rarely covers the full disease spectrum, rare cases, and diverse acquisition conditions, whereas privacy regulations, device heterogeneity, and institutional workflows make centralized raw-image collection impractical (Kaissis et al. 2020a; Bonawitz et al. 2019; Abadi et al. 2016). Therefore, integrating multicenter knowledge without exposing raw data is a practical and efficient route toward real clinical deployment.

Despite their success on natural images and general visual tasks (Izmailov et al. 2018; Wortsman et al. 2022; Ilharco et al. 2023; Entezari et al. 2022; Ainsworth, Hayase, and Srinivasa 2023), transferring these paradigms to medical data remains challenging. A key reason is that general-purpose model merging implicitly assumes that different client checkpoints contain relatively complete and comparable global discriminative structures; local models are expected to align through parameter space, task-vector space, update signs, or curvature statistics (Yadav et al. 2023; Yu et al. 2024; Matena and Raffel 2022; Jin et al. 2025). This assumption does not hold in medical scenarios. Medical centers differ not only in device domains but also in diagnostic-class availability: some centers may mainly cover common diseases or specific examination types, while having few rare-disease samples (Guan and Liu 2022; Zech et al. 2018; Li et al. 2021). The resulting local models are not complete replicas of the global task, but partial diagnostic experts supported by locally observed diseases.

As illustrated in Fig. 1, we summarize these challenges as two diagnostic-structure abnormalities in medical model merging:

- **Intra-client: limited local diagnostic coverage.** Each client has reliable supervision only for observed classes; thus, a local model may be locally discriminative but still lack evidence for the full global diagnostic space (Guan and Liu 2022; Li et al. 2021).
- **Inter-client: aggregation-induced prediction collapse.** Inconsistent class coverage and long-tailed prevalence can cause generic merging methods to mix reliable, absent, and conflicting class directions, making the merged global model concentrate predictions on a few dominant classes. More subtly, long-tailed medical distributions can make such collapse appear competitive under raw accuracy, thereby hiding insufficient global diagnostic discrimination (Buda, Maki, and Mazurowski 2018;

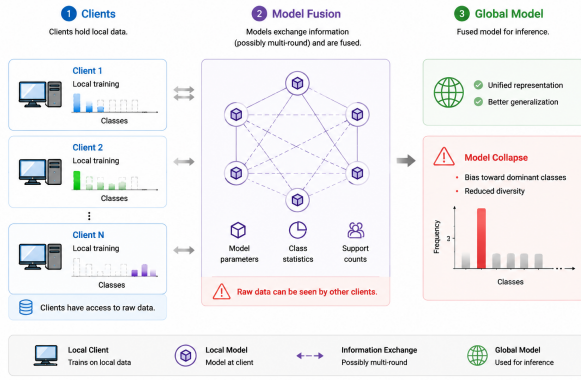


Figure 1: In multicenter healthcare settings, individual clients typically observe only a subset of the global diagnostic space. Conventional parameter-level model-merging methods can therefore cause the merged global model’s predictions to concentrate on a small number of dominant classes.

Brodersen et al. 2010).

To address these issues, we propose **LAMP-Merge**, a long-tail-aware medical prototype-merging method for single-shot asynchronous medical image model merging. LAMP-Merge shifts the merge target from global parameter interpolation to class-level diagnostic-evidence reconstruction. The key idea is to use client-uploaded class-level statistics as a synchronization signal for the global diagnostic space, allowing the server to reconstruct a global diagnostic classifier without accessing raw medical images, using a server-side validation set, or conducting multi-round federated communication. Specifically, LAMP-Merge consists of two mechanisms: diagnostic prototype reconstruction aggregates reliable class-level evidence through prototypes in a shared reference feature space, and long-tail prevalence calibration injects a bounded prior bias from client-uploaded class counts, thereby mitigating prediction collapse while preserving clinically meaningful majority-class prevalence.

Our main contributions are summarized as follows:

- **Medical Merging Diagnosis.** We revisit model merging under single-shot asynchronous medical collaboration and identify limited local diagnostic coverage and aggregation-induced prediction collapse as two key abnormalities caused by medical class availability and long-tailed prevalence.
- **LAMP-Merge.** We propose a long-tail-aware medical prototype-merging method that reconstructs a global diagnostic classifier from client-side class prototypes and prevalence statistics without exposing raw images or requiring multi-round federated optimization.
- **Empirical Validation.** We conduct full-scope experiments on five medical image datasets, four vision-model families, three client-population sizes, and three non-IID skew levels, showing that LAMP-Merge consistently outperforms strong model-merging baselines and im-

proves prediction-collapse diagnostics.

Related Works

Model Merging

Model-merging methods integrate independently trained checkpoints in parameter, task-vector, or geometric spaces. Weight averaging (Izmailov et al. 2018) and Model Soups (Wortsman et al. 2022) directly interpolate parameters; permutation-aware matching (Entezari et al. 2022; Ainsworth, Hayase, and Srinivasa 2023) aligns hidden-unit symmetries; ZipIt (Stoica et al. 2024) performs feature-level zipping; Task Arithmetic (Ilharco et al. 2023), TIES-Merging (Yadav et al. 2023), and DARE (Yu et al. 2024) manipulate task deltas and sign conflicts; Fisher merging (Matena and Raffel 2022), RegMean (Jin et al. 2025), Model Stock (Jang, Yun, and Han 2025), Bread-crumbs (Davari and Belilovsky 2024), Iso-C (Marczak et al. 2025), FreeMerge (Zheng and Wang 2025), and Robust-Merge (Zeng et al. 2025) introduce curvature, mean-statistic, subspace, frequency-domain, or robustness assumptions.

However, these methods share a class-agnostic merging interface: their variables are global parameters, task vectors, update signs, or geometric statistics rather than class-level diagnostic evidence. **They do not encode whether a client has reliable supervision for a specific medical class.** Under incomplete local class coverage and long-tailed prevalence, reliable class directions can be mixed with absent or conflicting directions from other clients, resulting in aggregation-induced prediction collapse (Buda, Maki, and Mazurowski 2018; Zech et al. 2018).

Medical Model Merging

Medical fusion studies exploit domain structures such as irregular clinical time series, physiological sensor graphs, and image-EHR fusion. mTAN (Shukla and Marlin 2021), Rain-drop (Zhang et al. 2022), and MedFuse (Hayat, Geras, and Shamout 2023) show that medical tasks benefit from modeling clinical sampling mechanisms, modality heterogeneity, and patient-level temporal relations.

However, their fusion objects are usually patient-level inputs, multimodal records, or clinical representations rather than independently trained image-classifier checkpoints. They often require patient samples, timestamps, electronic health records, or server-side validation feedback, which conflicts with common privacy constraints (Kaissis et al. 2020a; Sheller et al. 2020; Bonawitz et al. 2019). Med-Merge (Wei et al. 2026) is a relevant training-free medical knowledge-transfer framework, but it targets cross-modal adaptation for medical vision-language models rather than asynchronous checkpoint merging for the same classification task. **Thus, existing medical fusion routes are mismatched with our privacy and single-upload protocol constraints.** LAMP-Merge keeps the medical structure uploadable by using only locally computed class prototypes and prevalence counts.

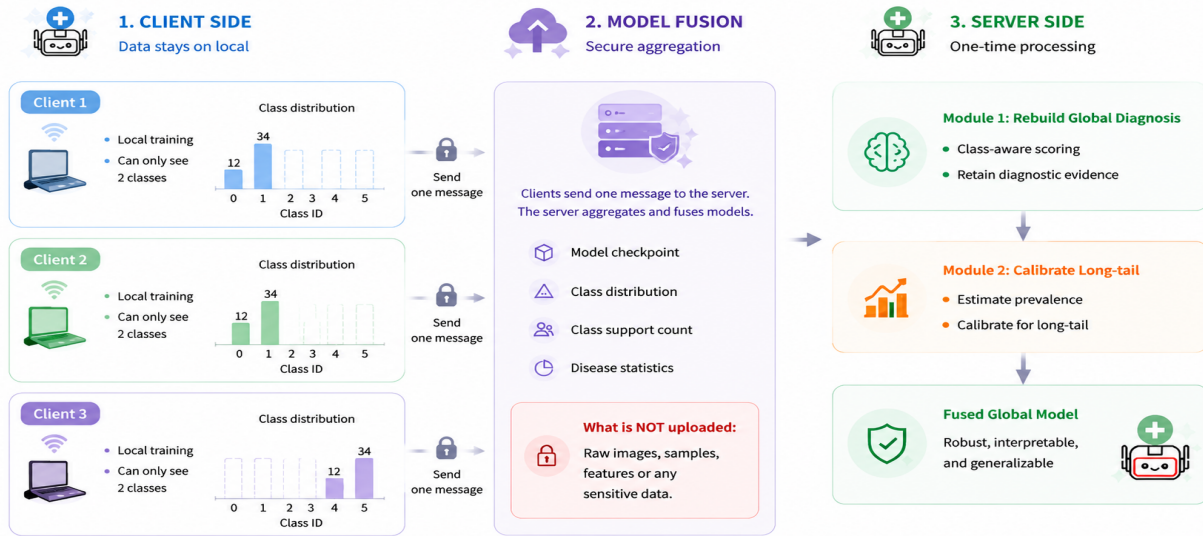


Figure 2: Local models from multicenter hospitals typically contain only partial diagnostic knowledge, while raw medical images cannot be centrally shared. LAMP-Merge reconstructs a privacy-friendly global medical classifier through diagnostic prototype reconstruction and long-tail prevalence calibration, without accessing raw data or requiring multi-round federated optimization.

Federated Learning

Federated and decentralized learning perform collaborative training through communication protocols. FedAvg (McMahan et al. 2017) and its variants or medical deployments (Li et al. 2020; Karimireddy et al. 2020; Li et al. 2021; Kaissis et al. 2020b; Sheller et al. 2020; Rieke et al. 2020; Kairouz et al. 2021) rely on repeated server-client updates; Swarm Learning (Wan et al. 2021) and Gossip Learning (Hegedüs, Danner, and Jelasity 2019) reduce centralization but still require participants to continue training, exchanging models, and optimizing from intermediate states.

However, our setting begins after local training has finished. Hospitals may not remain online, receive global checkpoints repeatedly, or run additional optimization rounds. **The bottleneck is therefore communication feasibility in addition to privacy.** LAMP-Merge requires one upload of a local model interface and class-level statistics, enabling the server to construct a unified diagnostic model without a multi-round feedback loop.

Method

In this section, we first formulate the single-shot asynchronous medical model-merging problem and characterize the diagnostic-structure pathologies exposed when generic model merging is applied to medical data, namely limited local diagnostic coverage and aggregation-induced prediction collapse. To correct these intrinsic mismatches, we propose **LAMP-Merge**, a two-stage framework composed of diagnostic prototype reconstruction and long-tail prevalence calibration, as shown in Fig. 2.

Preliminaries and Problem Formulation

Assume K medical clients. Client i owns a private labeled dataset $D_i = \{(x, y)\}$ drawn from a class-skewed medical distribution over C diagnosis classes. Let $D_{i,c}$ denote the subset of client i belonging to diagnosis class c . In medical partitions, $D_{i,c}$ can be empty for many classes; therefore, client i has reliable supervision only for its observed classes. We study single-shot asynchronous merging: after local training, each client uploads only once, while the server cannot access raw medical images, per-sample features, per-sample predictions, or a server-side validation set, and does not issue multi-round updates to clients. The goal is to construct a global diagnostic model under these constraints.

Clients are allowed to upload class-level statistics. Specifically, $n_{i,c}$ denotes the class-support count, $m_{i,c}$ denotes the class-prevalence count, and $\mu_{i,c}$ denotes the class prototype computed in a shared reference feature space. The server performs one merge using only the local checkpoint interfaces and these class-level statistics. This setting differs from federated learning because clients do not continue optimization from server feedback; it also differs from standard model merging because the presence of class-level diagnostic evidence becomes an explicit merging variable.

Motivation

Although generic model merging is training-free and parameter-efficient, we find that its diagnostic structure can severely degenerate in medical model merging. Prediction-distribution diagnosis and class-support analysis reveal two key abnormalities:

- **Limited local diagnostic coverage.** Each local checkpoint is a partial diagnostic expert: it can provide reliable evidence for classes present in D_i , but lacks supervised support for missing classes. Treating all local checkpoints as complete global-task replicas therefore introduces unsupported class directions into the merge.
- **Aggregation-induced prediction collapse.** Under partial class coverage, parameter-level merging can turn several non-degenerate local predictors into a globally collapsed predictor. The merged model tends to allocate most test images to a few dominant classes; because medical prevalence is often long-tailed, this collapse can still appear competitive in raw accuracy.

These observations suggest that medical model merging should not seek only an average point in the global parameter space. Instead, it should recover class-conditional diagnostic evidence and then preserve genuine long-tail prevalence. LAMP-Merge therefore shifts the merge target from parameter-space consensus to diagnosis-space evidence reconstruction.

Overview

The core framework of LAMP-Merge is illustrated in Fig. 2. The central idea is to use class-level client statistics as a synchronization signal for the global diagnostic space. This turns medical model merging from global parameter averaging into class-conditional diagnostic-evidence reconstruction, supplemented by calibrated class-prior information.

Notably, LAMP-Merge neither requires the server to access clients’ raw images nor relies on a server-side validation set, per-sample features, or per-sample predictions. Clients upload only class-level statistics; thus, the framework preserves the single-shot asynchronous boundary and differs from federated learning procedures that repeatedly broadcast models and retrain locally.

The framework contains two components. The first reconstructs multiclass diagnostic directions from class-supported prototypes, counteracting aggregation-induced collapse. The second injects a bounded long-tail prior, preventing the reconstruction step from overcorrecting clinically meaningful majority prevalence.

Diagnostic Prototype Reconstruction Module

Therefore, LAMP-Merge first reconstructs a class-wise diagnostic head in the shared reference feature space. Let ϕ_0 be the frozen reference backbone instantiated from the shared model configuration before local training, and let $T(\cdot)$ be the evaluation transform. For each diagnosis class c , client i computes a reference-space class prototype

$$\mu_{i,c} = \frac{1}{n_{i,c}} \sum_{(x,y) \in D_i} \mathbf{1}[y = c] \phi_0(T(x)), \quad (1)$$

where $n_{i,c} = |D_{i,c}|$ is the class support count. If $n_{i,c} = 0$, the client uploads zero support for that class and contributes no prototype evidence.

The server converts support counts into evidence weights:

$$e_{i,c} = (n_{i,c} + 1)^\gamma \mathbf{1}[n_{i,c} > 0], \quad (2)$$

where $\gamma = 0.45$ is the default evidence exponent. The normalized reliability of client i for class c is

$$\alpha_{i,c} = \frac{e_{i,c}}{\sum_{j=1}^K e_{j,c}}. \quad (3)$$

The global diagnosis prototype is then

$$p_c = \sum_{i=1}^K \alpha_{i,c} \mu_{i,c}. \quad (4)$$

Finally, LAMP-Merge synthesizes a cosine prototype classifier:

$$w_c = s \frac{p_c}{\|p_c\|_2}, \quad (5)$$

where $s = 20$ is the default head scale. This module addresses a key limitation of existing methods: it preserves the classes for which each client has supporting evidence and reconstructs a global multiclass diagnostic head.

Long-tail Prevalence Calibration

After obtaining this newly reconstructed diagnostic head, each client additionally uploads class-prevalence counts $m_{i,c}$ to account for the long-tailed prevalence inherent in real-world medical tasks. The server estimates the global class prior

$$\pi_c = \frac{\sum_{i=1}^K m_{i,c}}{\sum_{k=1}^C \sum_{i=1}^K m_{i,k}}. \quad (6)$$

We measure dominant-class imbalance by

$$r = C \max_c \pi_c. \quad (7)$$

The module is activated only when $r > \tau$, where $\tau = 2.5$ by default. When activated, it adds a centered log-prior bias

$$b_c = \lambda \left(\log \pi_c - \frac{1}{C} \sum_{k=1}^C \log \pi_k \right), \quad (8)$$

with default maximum calibration strength $\lambda = 5.0$. The final score is

$$\text{score}_c(x) = w_c^\top \phi_0(T(x)) + b_c. \quad (9)$$

The bias is centered, bounded, and attached to the prototype classifier; it calibrates prevalence without replacing the multi-class diagnostic directions reconstructed by the Diagnostic Prototype Reconstruction Module.

Experiments

Experimental Settings

1) Evaluation Datasets: We evaluate the merged models on five representative MedMNIST v2 benchmarks (Yang, Shi, and Ni 2023): Blood, Derma, Organ-C, Organ-S, and Ultrasound. They cover blood-cell microscopy, dermoscopy, abdominal CT, and ultrasound image classification (detailed dataset descriptions are provided in the Appendix).

2) Baselines: We compare LAMP-Merge with twelve model-merging baselines covering the main technical routes of current model merging: `avg`(Wortsman et al. 2022), `ties`(Yadav et al. 2023), `dare_linear`(Yu et al. 2024), `dare_ties`(Yu et al. 2024), `regmean`(Jin et al. 2025), `fisher`(Matena and Raffel 2022), `breadcrumbs`(Davari and Belilovsky 2024), `model_stock`(Jang, Yun, and Han 2025), `from`(Ilharco et al. 2023), `iso_c`(Marczak et al. 2025), `free_merge`(Zheng and Wang 2025), and `robustmerge`(Zeng et al. 2025). These baselines include weight averaging, task-vector pruning, sign-conflict resolution, curvature or mean-statistic correction, stock-style model interpolation, isotropic subspace merging, Fourier-domain fusion, and direction-robust merging.

3) Merging Settings: Experiments follow a single-shot asynchronous model-merging protocol. Each client independently trains a local model on its own medical imaging data and then computes the class-level statistics required by LAMP-Merge locally, including class-support counts, reference-space class prototypes, and class-prevalence counts. These statistics are uploaded once together with an interface to the local checkpoint. The server performs a single model-merging step using only the uploaded checkpoints and class-level statistics, and evaluates the resulting merged model on a global test set.

We evaluate four vision-model families: ResNet, ConvNeXt, ViT-Tiny, and Swin-Tiny(He et al. 2016; Liu et al. 2022; Dosovitskiy et al. 2021; Liu et al. 2021). For each dataset-model pair, we construct non-IID client partitions with $K \in \{3, 5, 7\}$ clients and Dirichlet skew parameter $\beta \in \{0, 0.01, 0.1\}$, using seed 42(Hsu, Qi, and Brown 2019). A client-average result fixes the dataset, model, and K , and then averages over the three β values.

4) Implementation Environment: All experiments were run on a server with two Intel(R) Xeon(R) Silver 4210 CPUs, four NVIDIA GeForce RTX 2080 Ti GPUs, and the conda environment MM with Python 3.10.20, PyTorch 2.6.0+cu118, and CUDA runtime 11.8.

Comparison With the State of the Arts

The main comparison uses client-average ACC: for each fixed dataset, model, and client number, we first average over the three β skew levels and then compare merging methods. A cell is counted as successful when LAMP-Merge is greater than or equal to the strongest non-LAMP baseline in that cell.

Tables 1–4 report the full client-average results for all four backbones. Across the 60 client-average cells, LAMP-Merge wins or ties the strongest non-LAMP baseline in **57 cells (95.0%)**.

Ablation Study

Module-level ablations use exactly the same experimental setup as the main experiments. We compare three configurations: full LAMP-Merge, diagnostic prototype reconstruction only, and weight averaging augmented only with long-tail prevalence calibration. Both components provide substantial performance improvements.

1) Effectiveness of Diagnostic Prototype Reconstruction

To further identify which information within diagnostic prototype reconstruction is necessary, we conduct controlled substitutions across the full setting. All configurations retain long-tail prevalence calibration and vary only the prototype semantics or support-count weighting.

Fig. 3 and Table 5 show that neither an arbitrary head nor a class-agnostic statistic explains the gain. Replacing reference-space prototypes or class-specific support weights consistently reduces ACC, confirming that **diagnostic semantics and class-level reliability are jointly necessary**.

2) Effectiveness of Long-tail Prevalence Calibration

We further replace the prior term inside long-tail prevalence calibration. This group changes only the class prior used by the calibration bias.

Table 6 shows that removing prevalence calibration or replacing the real prior with a uniform prior reduces mean ACC, confirming that long-tailed prevalence is part of the full method.

New-metric Diagnostic Experiment

Although the main experiments use ACC for direct comparison, long-tailed medical distributions can allow single-class or few-class predictors to achieve high ACC(Buda, Maki, and Mazurowski 2018; Brodersen et al. 2010). We therefore report balanced accuracy, macro-F1, collapse ratio, effective predicted classes, and predicted–true total variation.

Table 7 compares LAMP-Merge with the best-performing baseline for each metric. **LAMP-Merge consistently and substantially outperforms the existing methods**. This indicates that its accuracy gains do not arise from more severe single-class prediction collapse, but rather from more complete global diagnostic discrimination.

Visualization Experiment

To visualize the prediction-distribution changes measured by the new diagnostic metrics, we compare the predicted class allocation induced by LAMP-Merge with that of the strongest generic merge. The generic merge concentrates predictions on one class, whereas LAMP-Merge recovers a broader class allocation aligned with the test distribution.

Hyperparameter Analysis Experiment

Fig. 5 (top) evaluates the prototype-head scale s and calibration strength λ over 180 raw cells and 60 client-average cells. The default $s = 20$ and $\lambda = 5$ lie in broad stable regions, indicating that the gain is not produced by narrow hyperparameter tuning.

t-SNE Visualization Experiment

Fig. 5 (middle) uses t-SNE(van der Maaten and Hinton 2008) only for analysis on Blood with ResNet, $K = 3$, and $\beta = 0.01$. LAMP-Merge retains separated output-probability groups, while generic merges exhibit compressed or entangled predictive representations.

Table 1: Client-average test accuracy on ResNet. Each entry averages the three Dirichlet settings for a fixed client number; Avg denotes the mean over $K = 3, 5, 7$ within the corresponding dataset.

Method	Blood				Derma				Organ-C				Organ-S				Ultrasound			
	$K = 3$	$K = 5$	$K = 7$	Avg	$K = 3$	$K = 5$	$K = 7$	Avg	$K = 3$	$K = 5$	$K = 7$	Avg	$K = 3$	$K = 5$	$K = 7$	Avg	$K = 3$	$K = 5$	$K = 7$	Avg
Avg	0.3345	0.2467	0.2822	0.2878	0.6291	0.4700	0.6244	0.5745	0.2261	0.1772	0.1396	0.1810	0.2864	0.1794	0.2475	0.2378	0.2435	0.1581	0.1542	0.1853
TIES[NeurIPS 2023]	0.2601	0.1872	0.1787	0.2087	0.5995	0.4841	0.6484	0.5773	0.2650	0.1684	0.2118	0.2151	0.1981	0.1915	0.1874	0.1923	0.3477	0.1989	0.1854	0.2440
DARE-Linear[ICML 2024]	0.2832	0.2856	0.2392	0.2693	0.6357	0.4841	0.6288	0.5829	0.2018	0.2588	0.1650	0.2085	0.2545	0.2245	0.2537	0.2442	0.2162	0.1728	0.1608	0.1833
DARE-TIES[ICML 2024]	0.3194	0.1935	0.1780	0.2303	0.5375	0.4825	0.5558	0.5253	0.1851	0.1515	0.1859	0.1742	0.1790	0.1352	0.1730	0.1624	0.2932	0.1635	0.1890	0.2152
RegMean[ICLR 2023]	0.3460	0.2369	0.3407	0.3079	0.5812	0.2843	0.5022	0.4559	0.2405	0.1340	0.1388	0.1711	0.2504	0.1623	0.1493	0.1873	0.2276	0.1695	0.1318	0.1763
Fisher[NeurIPS 2022]	0.3196	0.2237	0.2608	0.2680	0.6707	0.6702	0.6690	0.6700	0.2973	0.1741	0.1812	0.2175	0.2010	0.2956	0.1963	0.2310	0.2333	0.1860	0.2513	0.2235
Breadcrumbs[ECCV 2024]	0.0854	0.1417	0.1460	0.1244	0.0495	0.2913	0.1721	0.1710	0.0766	0.0676	0.0685	0.0709	0.0802	0.0948	0.0603	0.0784	0.1225	0.1273	0.1021	0.1173
Model Stock[ECCV 2024]	0.2420	0.1824	0.1975	0.2073	0.6216	0.4311	0.5721	0.5416	0.1225	0.1274	0.1185	0.1228	0.2742	0.1578	0.2012	0.2111	0.1791	0.1321	0.1578	0.1563
FROM[ICLR 2023]	0.2989	0.2306	0.2052	0.2449	0.6713	0.4369	0.6580	0.5887	0.2350	0.1560	0.1520	0.1810	0.2207	0.2087	0.2476	0.2257	0.2309	0.1506	0.2063	0.1959
Iso-C[ICML 2025]	0.3278	0.2748	0.3023	0.3016	0.5049	0.4422	0.6416	0.5296	0.2331	0.1629	0.1383	0.1781	0.2736	0.2118	0.2453	0.2436	0.2615	0.1698	0.1554	0.1956
Free-Merging[ICCV 2025]	0.3345	0.2467	0.2822	0.2878	0.6291	0.4700	0.6246	0.5746	0.2261	0.1771	0.1396	0.1809	0.2863	0.1793	0.2475	0.2377	0.2435	0.1581	0.1545	0.1854
RobustMerge[NeurIPS 2025]	0.3401	0.2430	0.2256	0.2696	0.2956	0.2085	0.3942	0.2994	0.2458	0.1764	0.1653	0.1958	0.2680	0.2227	0.2301	0.2403	0.2189	0.2231	0.1707	0.2042
LAMP-Merge	0.8002	0.7999	0.8007	0.8003	0.6745	0.6761	0.6763	0.6756	0.6074	0.6069	0.6076	0.6073	0.5531	0.5524	0.5517	0.5524	0.4762	0.4762	0.4762	0.4762
	+ 0.47	+ 0.55	+ 0.52	+ 0.51	+ 0.05	+ 0.21	+ 0.05	+ 0.10	+ 0.38	+ 0.43	+ 0.47	+ 0.43	+ 0.27	+ 0.37	+ 0.30	+ 0.31	+ 0.23	+ 0.32	+ 0.32	+ 0.29

Table 2: Client-average test accuracy on ConvNeXt. Each entry averages the three Dirichlet settings for a fixed client number; Avg denotes the mean over $K = 3, 5, 7$ within the corresponding dataset.

Method	Blood				Derma				Organ-C				Organ-S				Ultrasound			
	$K = 3$	$K = 5$	$K = 7$	Avg	$K = 3$	$K = 5$	$K = 7$	Avg	$K = 3$	$K = 5$	$K = 7$	Avg	$K = 3$	$K = 5$	$K = 7$	Avg	$K = 3$	$K = 5$	$K = 7$	Avg
Avg	0.1565	0.1122	0.1384	0.1357	0.2966	0.4830	0.6688	0.4828	0.1037	0.1218	0.0944	0.1066	0.1308	0.1035	0.1416	0.1253	0.1506	0.1506	0.1252	0.1421
TIES[NeurIPS 2023]	0.1715	0.1214	0.1524	0.1484	0.2966	0.1797	0.2713	0.2492	0.1217	0.0786	0.0716	0.0906	0.1123	0.0761	0.0796	0.0893	0.1518	0.1456	0.1291	0.1422
DARE-Linear[ICML 2024]	0.1756	0.1122	0.1345	0.1408	0.2705	0.4830	0.4830	0.4122	0.1035	0.0699	0.1085	0.0940	0.1282	0.1035	0.0797	0.1038	0.1575	0.1575	0.1264	0.1471
DARE-TIES[ICML 2024]	0.1096	0.1519	0.1497	0.1371	0.0908	0.2798	0.2451	0.2052	0.1305	0.1346	0.0770	0.1140	0.1096	0.0946	0.0709	0.0917	0.1456	0.1297	0.1500	0.1418
RegMean[ICLR 2023]	0.1565	0.0817	0.1856	0.1413	0.2633	0.0773	0.2720	0.2042	0.1143	0.1017	0.0768	0.0976	0.1014	0.1259	0.1038	0.1104	0.1575	0.1539	0.1500	0.1538
Fisher[NeurIPS 2022]	0.1565	0.0999	0.1189	0.1251	0.4825	0.2638	0.2971	0.3478	0.1003	0.0974	0.0915	0.0964	0.0604	0.0517	0.0843	0.0655	0.1695	0.1360	0.1456	0.1504
Breadcrumbs[ECCV 2024]	0.1516	0.1366	0.1163	0.1348	0.0524	0.2766	0.4630	0.2640	0.1031	0.1444	0.1574	0.1350	0.1516	0.1554	0.2077	0.1716	0.1384	0.1363	0.1363	0.1370
Model Stock[ECCV 2024]	0.1756	0.1144	0.1778	0.1559	0.2966	0.4830	0.6688	0.4828	0.1276	0.1218	0.1385	0.1293	0.1308	0.1035	0.1935	0.1426	0.1575	0.1291	0.1479	0.1448
FROM[ICLR 2023]	0.1565	0.1083	0.1671	0.1440	0.2966	0.3406	0.4569	0.3647	0.1232	0.1224	0.0650	0.1035	0.1324	0.1105	0.0897	0.1109	0.1575	0.1578	0.1087	0.1413
Iso-C[ICML 2025]	0.1947	0.1865	0.1480	0.1764	0.2971	0.4830	0.6688	0.4830	0.0991	0.0699	0.0945	0.0878	0.0789	0.1477	0.0867	0.1044	0.1518	0.1291	0.1605	0.1471
Free-Merging[ICCV 2025]	0.1575	0.1190	0.1384	0.1383	0.2971	0.4830	0.6688	0.4830	0.0906	0.1218	0.1385	0.1170	0.1308	0.1035	0.1416	0.1253	0.1506	0.1291	0.1695	0.1497
RobustMerge[NeurIPS 2025]	0.1410	0.1405	0.0752	0.1189	0.0908	0.4507	0.4830	0.3415	0.1394	0.1218	0.1793	0.1468	0.2354	0.0961	0.2077	0.1797	0.1300	0.1135	0.1363	0.1266
LAMP-Merge	0.8482	0.8483	0.8493	0.8486	0.5992	0.5920	0.5852	0.5921	0.6482	0.6480	0.6495	0.6486	0.5977	0.5985	0.5990	0.5984	0.4259	0.4259	0.4259	0.4259
	+ 0.69	+ 0.74	+ 0.71	+ 0.71	+ 0.30	+ 0.11	+ 0.08	+ 0.11	+ 0.54	+ 0.53	+ 0.56	+ 0.54	+ 0.47	+ 0.50	+ 0.46	+ 0.47	+ 0.28	+ 0.28	+ 0.30	+ 0.28

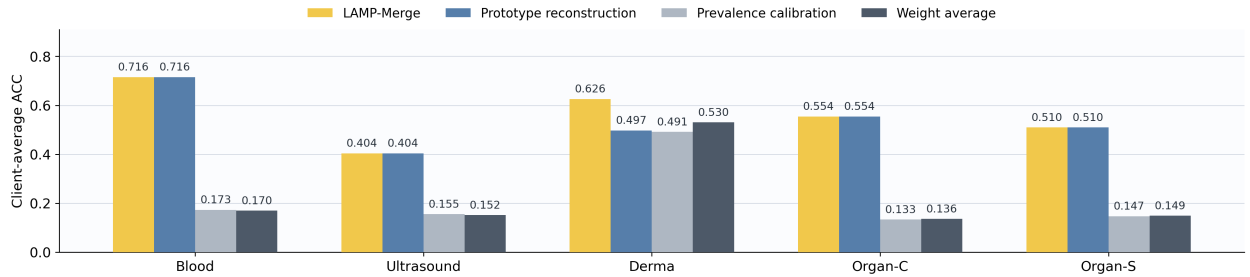


Figure 3: Dataset-level client-average accuracy in the module ablation. Each dataset contains 12 cells from four vision-model families and three client counts; each cell is averaged over three Dirichlet skew levels.

In-depth Analysis Experiment

Fig. 5 (bottom) summarizes inter-class separation, nearest-class margin, within-class client consistency, and prototype-client alignment. Together with Table 5, the joint profile confirms that class-semantic reference-space prototypes and

class-specific support weights provide the mechanism behind the full-scope gain.

Table 3: Client-average test accuracy on ViT-Tiny. Each entry averages the three Dirichlet settings for a fixed client number; Avg denotes the mean over $K = 3, 5, 7$ within the corresponding dataset.

Method	Blood				Derma				Organ-C				Organ-S				Ultrasound			
	$K = 3$	$K = 5$	$K = 7$	Avg	$K = 3$	$K = 5$	$K = 7$	Avg	$K = 3$	$K = 5$	$K = 7$	Avg	$K = 3$	$K = 5$	$K = 7$	Avg	$K = 3$	$K = 5$	$K = 7$	Avg
Avg	0.1343	0.1082	0.1160	0.1195	0.4638	0.4369	0.3696	0.4234	0.1657	0.1043	0.0952	0.1217	0.0844	0.0852	0.1710	0.1135	0.1450	0.1327	0.1884	0.1554
TIES[NeurIPS 2023]	0.0796	0.1421	0.1285	0.1167	0.4780	0.3498	0.4615	0.4298	0.1341	0.1223	0.0953	0.1172	0.1192	0.1002	0.1597	0.1264	0.2001	0.1306	0.1018	0.1442
DARE-Linear[ICML 2024]	0.2397	0.1129	0.1387	0.1638	0.2850	0.1002	0.1204	0.1685	0.1534	0.0946	0.0704	0.1061	0.0748	0.0549	0.1611	0.0969	0.1366	0.1051	0.1435	0.1284
DARE-TIES[ICML 2024]	0.0735	0.1189	0.1146	0.1023	0.2735	0.4502	0.3528	0.3588	0.1252	0.0903	0.1033	0.1063	0.0541	0.0874	0.0591	0.0669	0.1447	0.1120	0.1042	0.1203
RegMean[ICLR 2023]	0.1748	0.1417	0.1290	0.1485	0.5114	0.3667	0.6658	0.5146	0.1250	0.0964	0.0983	0.1066	0.1263	0.0717	0.1068	0.1016	0.1491	0.1384	0.1144	0.1340
Fisher[NeurIPS 2022]	0.1166	0.1512	0.1944	0.1541	0.4780	0.2042	0.4820	0.3881	0.0799	0.1072	0.1424	0.1098	0.0856	0.1218	0.1222	0.1099	0.1611	0.1315	0.1420	0.1449
Breadcrumbs[ECCV 2024]	0.1363	0.1039	0.1300	0.1234	0.0489	0.2572	0.0607	0.1223	0.0773	0.1000	0.1644	0.1139	0.1237	0.0483	0.1219	0.0980	0.1360	0.1141	0.1524	0.1342
Model Stock[ECCV 2024]	0.1940	0.1120	0.1134	0.1398	0.3805	0.3079	0.2735	0.3206	0.1488	0.1227	0.1215	0.1310	0.0893	0.0859	0.1867	0.1206	0.1599	0.1809	0.1641	0.1683
FROM[ICLR 2023]	0.1802	0.0781	0.1405	0.1329	0.4695	0.5121	0.6688	0.5501	0.2148	0.1301	0.0972	0.1474	0.0772	0.0798	0.1485	0.1018	0.1836	0.1743	0.1776	0.1785
Iso-C[ICML 2025]	0.1234	0.1483	0.1018	0.1245	0.5709	0.4529	0.2306	0.4181	0.1084	0.0567	0.0897	0.0849	0.0611	0.0531	0.1081	0.0741	0.1417	0.1527	0.1869	0.1604
Free-Merging[ICCV 2025]	0.1989	0.1457	0.1392	0.1613	0.4630	0.4770	0.3855	0.4418	0.1425	0.0980	0.0759	0.1055	0.0727	0.0914	0.1643	0.1095	0.1405	0.1593	0.1647	0.1548
RobustMerge[NeurIPS 2025]	0.1482	0.1070	0.1425	0.1326	0.0347	0.2768	0.1724	0.1613	0.0954	0.0617	0.1438	0.1003	0.1260	0.0707	0.1217	0.1061	0.1698	0.1854	0.1563	0.1705
LAMP-Merge	0.8523	0.8522	0.8516	0.8520	0.5970	0.5938	0.5860	0.5923	0.5805	0.5790	0.5790	0.5795	0.5406	0.5372	0.5388	0.5389	0.4645	0.4645	0.4645	0.4645
	+ 0.72	+ 0.74	+ 0.74	+ 0.73	+ 0.13	+ 0.16	+ 0.22	+ 0.17	+ 0.41	+ 0.47	+ 0.48	+ 0.46	+ 0.46	+ 0.45	+ 0.37	+ 0.43	+ 0.32	+ 0.33	+ 0.28	+ 0.31

Table 4: Client-average test accuracy on Swin-Tiny. Each entry averages the three Dirichlet settings for a fixed client number; Avg denotes the mean over $K = 3, 5, 7$ within the corresponding dataset.

Method	Blood				Derma				Organ-C				Organ-S				Ultrasound			
	$K = 3$	$K = 5$	$K = 7$	Avg	$K = 3$	$K = 5$	$K = 7$	Avg	$K = 3$	$K = 5$	$K = 7$	Avg	$K = 3$	$K = 5$	$K = 7$	Avg	$K = 3$	$K = 5$	$K = 7$	Avg
Avg	0.1565	0.1186	0.1524	0.1425	0.4569	0.4830	0.6688	0.5362	0.1068	0.1294	0.0787	0.1050	0.0682	0.0604	0.1381	0.0889	0.1527	0.1294	0.1465	0.1429
TIES[NeurIPS 2023]	0.0972	0.1190	0.1384	0.1182	0.0841	0.0780	0.6688	0.2770	0.1160	0.0702	0.0934	0.0932	0.0734	0.0607	0.1107	0.0816	0.1695	0.1303	0.1261	0.1420
DARE-Linear[ICML 2024]	0.1565	0.1161	0.1825	0.1517	0.4569	0.2771	0.4630	0.3990	0.1155	0.1272	0.0776	0.1068	0.0640	0.0720	0.1381	0.0914	0.1743	0.1536	0.1497	0.1592
DARE-TIES[ICML 2024]	0.1384	0.1162	0.1384	0.1310	0.2961	0.2638	0.6688	0.4096	0.1276	0.0774	0.1116	0.1055	0.1194	0.0434	0.1230	0.0953	0.1791	0.1351	0.1300	0.1481
RegMean[ICLR 2023]	0.1374	0.1715	0.1578	0.1556	0.2705	0.4630	0.4569	0.3968	0.1411	0.0773	0.0811	0.0998	0.1215	0.0821	0.0926	0.0987	0.1465	0.1105	0.1779	0.1450
Fisher[NeurIPS 2022]	0.1633	0.1756	0.1565	0.1651	0.4582	0.0652	0.4718	0.3317	0.0814	0.0899	0.0978	0.0897	0.0651	0.0650	0.1092	0.0798	0.1746	0.1165	0.1755	0.1555
Breadcrumbs[ECCV 2024]	0.1500	0.1365	0.1537	0.1467	0.0647	0.2771	0.0715	0.1378	0.1445	0.1783	0.1660	0.1629	0.2354	0.2077	0.2111	0.2181	0.1450	0.1312	0.1653	0.1472
Model Stock[ECCV 2024]	0.1374	0.1440	0.1715	0.1510	0.4569	0.2771	0.4825	0.4055	0.1110	0.0771	0.1404	0.1095	0.0671	0.1106	0.1714	0.1164	0.1539	0.1273	0.1869	0.1560
FROM[ICLR 2023]	0.1565	0.1495	0.1671	0.1577	0.4569	0.2971	0.6688	0.4743	0.1174	0.1220	0.1051	0.1148	0.1133	0.1016	0.1380	0.1176	0.1539	0.1252	0.1476	0.1422
Iso-C[ICML 2025]	0.1430	0.1722	0.1700	0.1617	0.3099	0.2840	0.5174	0.3704	0.1302	0.1327	0.1469	0.1366	0.1545	0.1727	0.1880	0.1717	0.1755	0.1731	0.1791	0.1759
Free-Merging[ICCV 2025]	0.1154	0.0861	0.1840	0.1285	0.4569	0.4830	0.6688	0.5362	0.0959	0.1377	0.0890	0.1075	0.0682	0.0497	0.1062	0.0747	0.1518	0.1575	0.1599	0.1564
RobustMerge[NeurIPS 2025]	0.1374	0.1409	0.1670	0.1484	0.2377	0.2771	0.0504	0.1884	0.1601	0.1442	0.1960	0.1668	0.1673	0.1151	0.2077	0.1634	0.1761	0.1315	0.1629	0.1568
LAMP-Merge	0.7737	0.7753	0.7748	0.7746	0.6374	0.6251	0.6244	0.6290	0.6651	0.6632	0.6630	0.6638	0.6078	0.6076	0.6068	0.6074	0.4816	0.4816	0.4816	0.4816
	+ 0.62	+ 0.66	+ 0.62	+ 0.63	+ 0.18	+ 0.14	+ 0.04	+ 0.09	+ 0.56	+ 0.53	+ 0.58	+ 0.56	+ 0.54	+ 0.55	+ 0.47	+ 0.52	+ 0.33	+ 0.35	+ 0.34	+ 0.34

Conclusion

We studied **one-shot asynchronous medical model merging**, where independently trained hospital models must be fused without federated optimization rounds and without exposing raw medical images to the server. Our empirical analysis shows that generic model-merging methods suffer from **aggregation-induced predictive collapse** under incomplete local diagnostic coverage. We further show that, on long-tailed medical datasets, such collapse can be masked by raw accuracy. We therefore introduce the Long-tail Prevalence Calibration module and perform secondary evaluation using macro recall, macro-F1, and predictive-distribution diagnostics (Buda, Maki, and Mazurowski 2018; Brodersen et al. 2010).

Taken together, the full-scope comparisons, prediction-distribution diagnostics, and prototype-error analysis establish a consistent mechanism: class-level support-weighted evidence reconstructs discriminative directions for locally

absent diagnoses, while bounded prevalence calibration retains clinically meaningful imbalance without reintroducing majority-class collapse.

LAMP-Merge addresses these two properties through a concise two-module design. Future work will extend its applicability in two directions: first, from medical image classification to segmentation, detection, and multimodal diagnostic tasks; and second, toward stronger privacy-preserving mechanisms, such as differential privacy and secure aggregation, to further reduce potential privacy risks associated with class-level statistics (Kaissis et al. 2020a; Geyer, Klein, and Nabi 2017; Abadi et al. 2016). We will also explore deployment and validation in larger-scale real-world multicenter healthcare settings.

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Table 5: Internal ablation of diagnostic prototype reconstruction. All settings cover 180 raw cells and 60 client-average cells.

Setting	Mean Acc	Δ vs. LAMP
LAMP-Merge	0.6209	–
Classifier-head aggregation	0.2579	↓36.30%
Global-feature mean	0.2645	↓35.64%
Support-only synthetic head	0.1137	↓50.73%
Shuffled-label prototype	0.2002	↓42.07%
Uniform client weight	0.5839	↓3.70%
Binary support only	0.5517	↓6.92%
Global client-size weight	0.5944	↓2.66%

Table 6: Internal ablation of long-tail prevalence calibration.

Setting	Mean Acc	Δ vs. LAMP
LAMP-Merge	0.6209	–
No prevalence calibration	0.5879	↓3.30%
Uniform prevalence prior	0.5879	↓3.30%

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Table 7: Prediction-collapse diagnostics averaged over five medical datasets and four backbones per dataset.

Metric	LAMP-Merge	Best reference	Δ vs. reference
Accuracy $\uparrow$	0.6211	0.2228	$\uparrow$ +39.83%
BA / macro recall $\uparrow$	0.5370	0.1492	$\uparrow$ +38.78%
Macro F1 $\uparrow$	0.5238	0.0757	$\uparrow$ +44.81%
Collapse ratio $\downarrow$	0.3122	0.7902	↓-47.80%
Effective classes $\uparrow$	7.1322	2.0489	$\uparrow$ +5.0833
Pred-true TV $\downarrow$	0.1262	0.7040	↓-57.78%

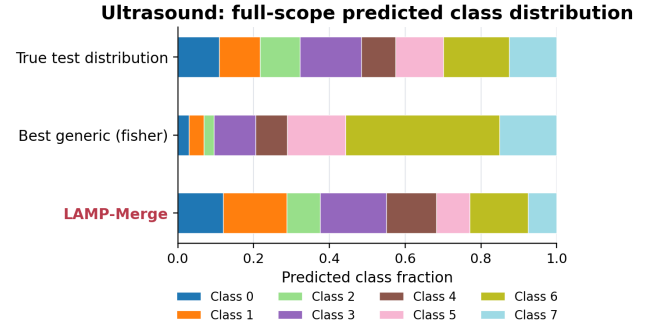


Figure 4: Predicted class distributions on Ultrasound, averaged over the four visual backbones, $K \in \{3, 5, 7\}$, and the three Dirichlet skew levels. The generic merge concentrates predictions on one class, while LAMP-Merge recovers a broader class allocation aligned with the test distribution.

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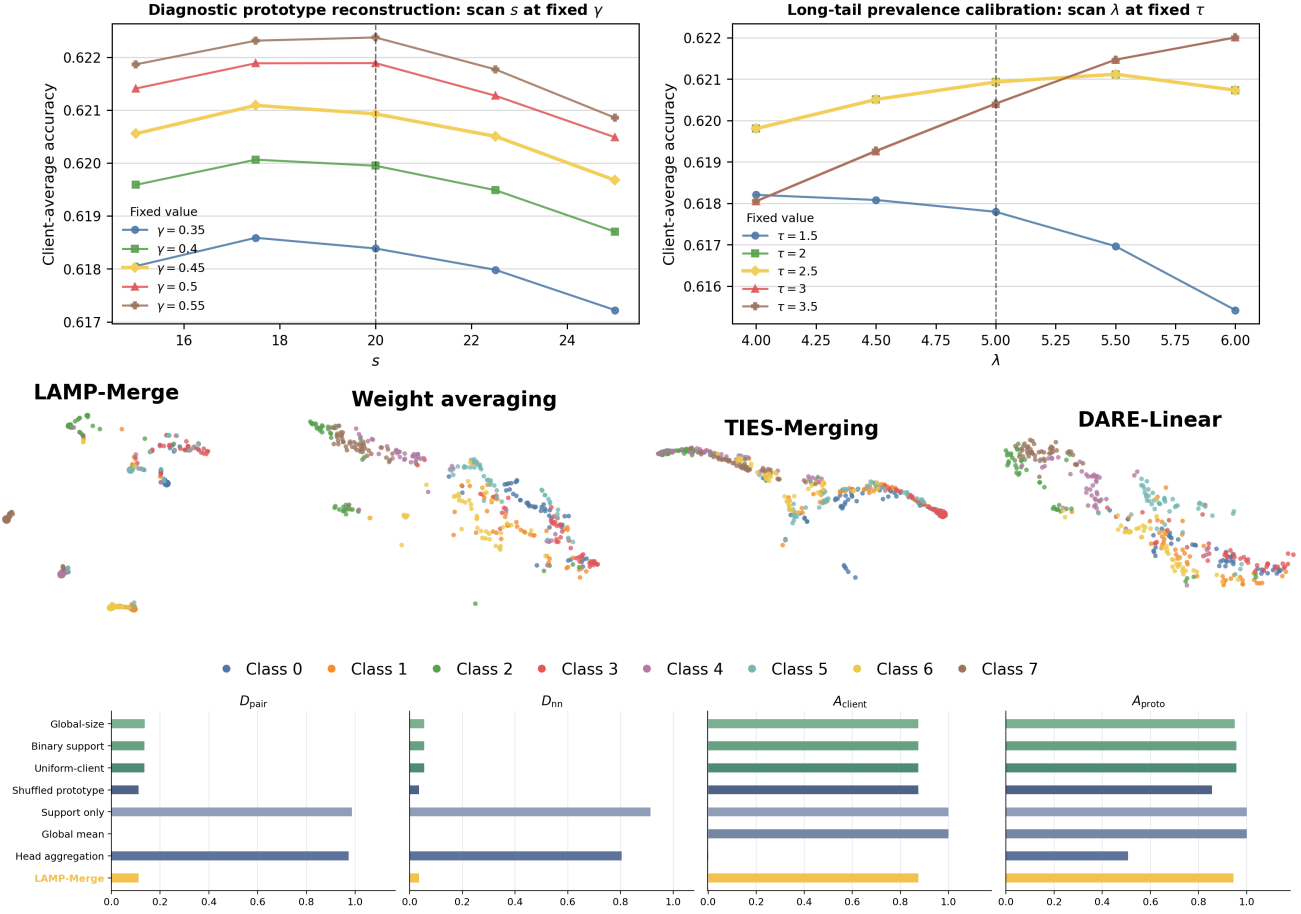


Figure 5: Core mechanism analyses. Top: full-scope hyperparameter sensitivity, where $s = 20$ and $\lambda = 5$ lie in stable high-accuracy regions across 180 raw cells and 60 client-average cells. Middle: joint t-SNE of test-image output-probability representations on Blood; from left to right, LAMP-Merge, weight averaging, TIES-Merging, and DARE-Linear. Bottom: full-scope prototype geometry, summarizing pairwise distance, nearest-class distance, prototype consistency, and prototype–client alignment across controlled prototype substitutions. The three panels jointly show that the performance gain is stable, preserves separated multi-class predictions, and is supported by class-semantic prototype geometry rather than an arbitrary classifier head.

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Supplementary Results and Analysis

Evaluation Datasets

The five MedMNIST v2 benchmarks (Yang, Shi, and Ni 2023) cover blood-cell microscopy, dermoscopy, abdominal CT organ slices, and ultrasound images. They involve cell morphology recognition, skin-lesion diagnosis, organ-structure recognition, and ultrasound-structure discrimination:

- **Blood:** an eight-class blood-cell microscopy dataset for hematological morphology classification, focusing on fine-grained cellular appearance and inter-class morphological similarity.
- **Derma:** a seven-class dermoscopic skin-lesion dataset, characterized by strong long-tailed prevalence and visually subtle lesion categories.
- **Organ-C:** an 11-class abdominal CT dataset in the coronal view, evaluating organ-category recognition under grayscale cross-sectional imaging.
- **Organ-S:** an 11-class abdominal CT dataset in the sagittal view, complementing Organ-C with a different anatomical imaging plane.
- **Ultrasound:** an eight-class ultrasound image dataset, covering structure recognition under modality-specific speckle noise and low-contrast boundaries.

Together, these benchmarks provide a systematic testbed for evaluating cross-modality robustness, class-discriminative capability, and task generalization under different medical image types and class-distribution conditions.

Detailed Ablation Configurations

All diagnostic-prototype substitutions retain long-tail prevalence calibration and vary only prototype semantics or support-count weighting:



Figure 6: Representative training images from the five medical image datasets used in the experiments. The class IDs are shown only to illustrate category diversity within each dataset.

- **LAMP-Merge** uses $p_c = \sum_i \alpha_{i,c} \mu_{i,c}$, where $e_{i,c} = (n_{i,c} + 1)^\gamma \mathbf{1}[n_{i,c} > 0]$ determines the class-support reliability.
- **Classifier-head aggregation** replaces the reference-space class prototype by the local classifier-head direction, i.e., $\mu_{i,c} \leftarrow h_{i,c}$.
- **Global-feature mean** removes class conditioning by setting $\mu_{i,c} \leftarrow g$ for all c , where g is the class-agnostic global reference-feature mean.
- **Support-only synthetic head** removes image evidence by replacing each prototype with a fixed random unit direction u_c , i.e., $p_c \leftarrow u_c$, $\|u_c\|_2 = 1$.
- **Shuffled-label prototype** preserves prototype magnitudes but permutes diagnostic semantics, i.e., $p_c \leftarrow p_{\sigma(c)}$, where σ is a random permutation over diagnosis labels.
- **Uniform client weight** replaces support-weighted aggregation with equal weights among clients containing class c , i.e., $\alpha_{i,c} \leftarrow \mathbf{1}[n_{i,c} > 0] / \sum_j \mathbf{1}[n_{j,c} > 0]$.
- **Binary support only** replaces fine-grained support and prevalence counts with class-presence indicators, i.e., $n_{i,c}, m_{i,c} \leftarrow \mathbf{1}[n_{i,c} > 0]$.
- **Global client-size weight** uses the total client support $N_i = \sum_k n_{i,k}$ rather than class-specific support, i.e., $e_{i,c} \leftarrow (N_i + 1)^\gamma \mathbf{1}[n_{i,c} > 0]$.

The long-tail prevalence substitutions vary only the class prior used by the calibration bias:

- **LAMP-Merge** estimates the empirical class prior as $\pi_c = \sum_i m_{i,c} / \sum_k \sum_i m_{i,k}$.
- **No prevalence calibration** removes the calibration bias by setting $b_c \leftarrow 0$.
- **Uniform prevalence prior** removes long-tail prevalence by setting $\pi_c \leftarrow 1/C$, where C is the number of diagnosis classes.

2) Additional Theoretical Analysis Let μ_c^* denote the true class mean in the shared reference space. If client class features have bounded second moments, the prototype estimation error satisfies

$$\mathbb{E} \|p_c - \mu_c^*\|_2^2 \leq \sum_i \alpha_{i,c}^2 \frac{\sigma_c^2}{n_{i,c}} + \text{Bias}_c^2. \quad (10)$$

This upper bound shows that support counts directly control the variance of class-prototype estimation. A client that does not observe class c should not contribute evidence for that class, and a client with more samples should contribute more, but only sublinearly.

Let the true margin between classes c and d for sample x be

$$\Delta_{c,d}(x) = (\mu_c^*)^\top z - (\mu_d^*)^\top z. \quad (11)$$

When the prototype error satisfies

$$\|p_c - \mu_c^*\|_2 + \|p_d - \mu_d^*\|_2 < \frac{\Delta_{c,d}(x)}{\|z\|_2}, \quad (12)$$

the estimated prototypes preserve the class order between c and d .

The role of long-tail prevalence calibration is to introduce long-tail priors in a bounded manner. For prevalence counts $m_{i,c}$, the class prior is

$$\pi_c = \frac{\sum_i m_{i,c}}{\sum_k \sum_i m_{i,k}}, \quad (13)$$

and the centered log-prior bias is

$$b_c = \lambda \left(\log \pi_c - \frac{1}{C} \sum_k \log \pi_k \right). \quad (14)$$

As long as λ is bounded, long-tail prevalence calibration does not replace the discriminative directions reconstructed by diagnostic prototype reconstruction; it only calibrates the output toward the real class prevalence. This explains why using long-tail prevalence calibration alone is insufficient, while full LAMP-Merge remains both non-collapsed and prevalence-aware.

Metric Definitions and Geometry Details

Let y_n and $\hat{y}_n$ denote the true and predicted labels of the n -th test sample. Let $q_c = N^{-1} \sum_n \mathbf{1}[\hat{y}_n = c]$ be the predicted-class distribution, $p_c = N^{-1} \sum_n \mathbf{1}[y_n = c]$ the true class distribution, and $\mathcal{C}_{\text{test}} = \{c \mid \sum_n \mathbf{1}[y_n = c] > 0\}$ the set of classes observed in the test set. We use:

- **BA** = $|\mathcal{C}_{\text{test}}|^{-1} \sum_{c \in \mathcal{C}_{\text{test}}} \text{Recall}_c$, balanced accuracy or macro recall, which measures whether all observed diagnosis classes are recalled.

- $\text{MacroF1} = |\mathcal{C}_{\text{test}}|^{-1} \sum_{c \in \mathcal{C}_{\text{test}}} \text{F1}_c$, class-balanced precision-recall quality that penalizes majority-only predictions.
- $\rho = \max_c q_c$, the prediction-collapse ratio, which measures concentration on a single diagnosis class.
- $C_{\text{eff}} = \exp(-\sum_c q_c \log q_c)$, the effective number of predicted classes.
- $\text{TV}(q, p) = \frac{1}{2} \sum_c |q_c - p_c|$, the total variation distance between predicted and true class distributions.

Let $d(a, b) = 1 - \cos(a, b)$ denote cosine distance and let $\mathcal{I}_c = \{i \mid n_{i,c} > 0\}$ denote the clients that contain diagnosis class c . The geometric quantities are:

- $D_{\text{pair}} = \frac{2}{C(C-1)} \sum_{c < d} d(p_c, p_d)$, the mean distance between global prototypes of different diagnosis classes.
- $D_{\text{nn}} = \frac{1}{C} \sum_c \min_{d \neq c} d(p_c, p_d)$, the mean nearest-class distance and most vulnerable decision margin.
- $A_{\text{client}} = \frac{1}{C} \sum_c \text{Avg}_{i < j, i, j \in \mathcal{I}_c} \cos(\mu_{i,c}, \mu_{j,c})$, the within-class consistency among client prototypes;
 $A_{\text{proto}} = \frac{1}{C} \sum_c \sum_{i \in \mathcal{I}_c} \alpha_{i,c} \cos(p_c, \mu_{i,c})$, the alignment between the global prototype and contributing client prototypes.